📝 Unit 1 Guided Notes — The Real Number System

Name _____ Date _____ Period _____

Classify it, root it, estimate it, convert it, and order it. Every number has a home in the real number system.

🎯 By the end of this unit, I can…

- Sort numbers into the real number system (rational vs. irrational)
- Find perfect squares, perfect cubes, and their roots
- Estimate square roots on a number line and to the nearest tenth
- Convert between fractions and decimals, including repeating decimals
- Compare and order real numbers from least to greatest

Standards: NC.8.NS.1 · NC.8.NS.2

Fill in the blanks as we go. For every example and problem, **write each step on its own line** in the space given, and box your final answer.

1 Number Classification

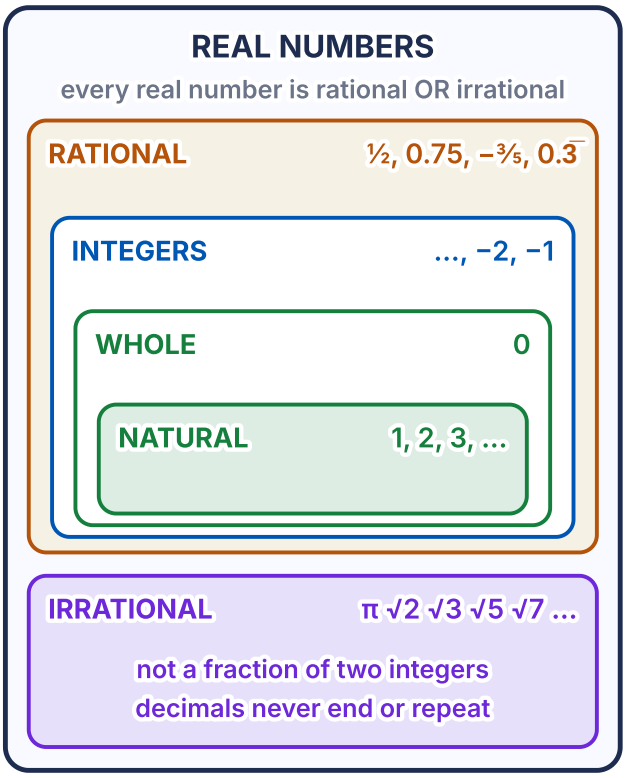
Every real number belongs to one or more of these sets. Fill in the definition for each:

Set	Examples	Definition
Natural Numbers	1, 2, 3, 4, …	_____ numbers
Whole Numbers	0, 1, 2, 3, …	Naturals + _____
Integers	…, −2, −1, 0, 1, 2, …	Whole numbers + their _____
Rational Numbers	$\frac{1}{2}$, 0.75, $-\frac{3}{5}$, $0.\overline{3}$	Can be written as a _____ of two integers (includes terminating & repeating decimals)

Irrational Numbers	π, √2, √3	CANNOT be written as a fraction of two integers; _____, _____ decimal

The Real Number System — Nested Sets

Each set fits *inside* the next one. 5 is natural, whole, an integer, and rational. π is not in any of these boxes.



Natural → Whole → Integers → Rational (each fits inside the next). Irrational numbers sit **outside** the rational box.

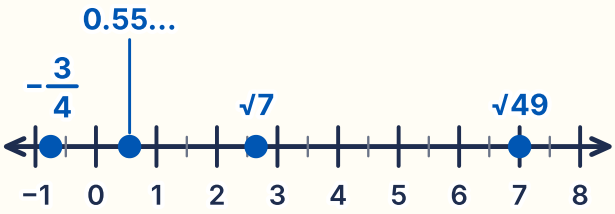
Match the Set to its Definition

MATCH **Click a set on the left, then its definition on the right.**

_____ 1. Natural Numbers	**A.** Whole numbers + their negatives
_____ 2. Whole Numbers	**B.** Counting numbers: 1, 2, 3, ...
_____ 3. Integers	**C.** Non-repeating, non-terminating decimal
_____ 4. Irrational Numbers	**D.** Naturals + zero

Quick Check

MC **Which number is *irrational*?**

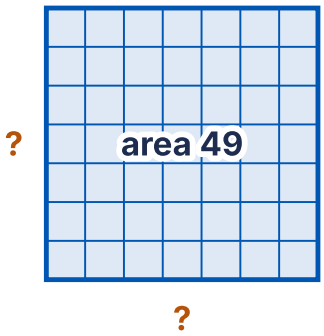


All four choices on a number line. Which one has a decimal that never ends **and** never repeats?

- ○ $0.5\overline{5}$
- ○ √49
- ○ √7
- ○ $-\frac{3}{4}$

ERROR ANALYSIS **A student claims:**

`"√49 is irrational because it has a square-root symbol."` **What's the mistake?**

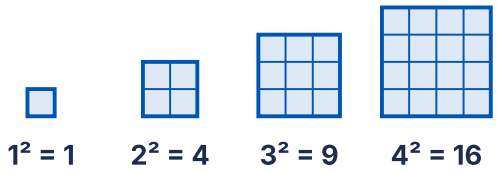


√49 asks: what side length makes a square with area 49? Count the tiles on one side.

2 Perfect Squares & Square Roots

📌 **THE RELATIONSHIP:** Squaring and square-rooting are INVERSE OPERATIONS — they undo each other. If $5^2 = 25$, then $\sqrt{25} = 5$. Know one, know the other!

A **perfect square** is a whole number times itself ($n \times n = n^2$). Its **square root** takes you back to that whole number. So $\sqrt{}$(a perfect square) is always a whole number.



Perfect squares are numbers of tiles that make a **perfect square**. The side is the square root: $\sqrt{16} = 4$.

Perfect Squares 1–15 — Reference Chart

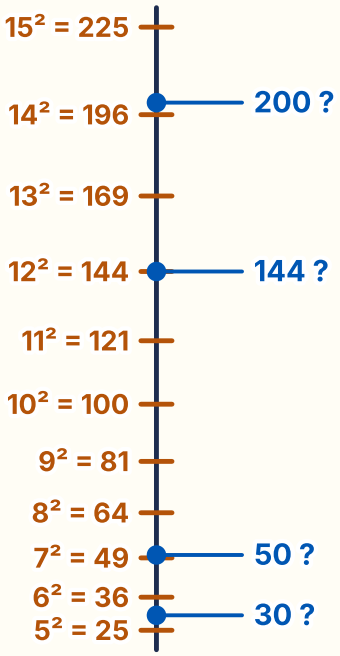
Keep this chart handy — you'll use it constantly for roots and estimating.

n	n^2	n	n^2	n	n^2
1	1	6	36	11	121
2	4	7	49	12	144
3	9	8	64	13	169
4	16	9	81	14	196
5	25	10	100	15	225

💡 **READ IT BOTH WAYS:** the chart says $12^2 = 144$, which means $\sqrt{144} = 12$. Every row is two facts in one.

Quick Check

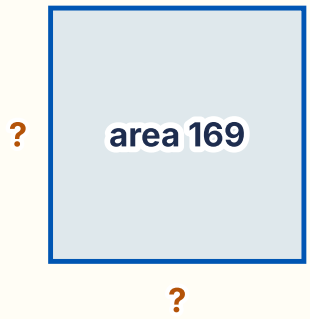
MC **Which of these is a *perfect square*?**



Perfect squares from 5^2 to 15^2. A perfect square lands **exactly on** one of the gold marks.

- ○ 30
- ○ 50
- ○ 144
- ○ 200

CALC **What is √169 ?**



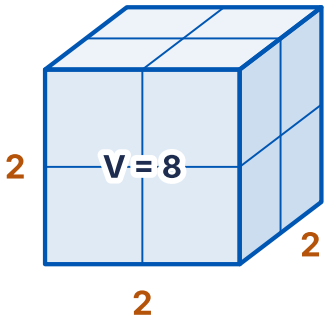
What side length makes a square with area 169?

👉 **WHAT IF IT'S NOT A PERFECT SQUARE?** Then the root is irrational (like √7 or √20) — it never ends or repeats. We estimate those in Section 4.

3 Perfect Cubes & Cube Roots

📌 **THE RELATIONSHIP:** Cubing and cube-rooting are INVERSE OPERATIONS too. If $5^3 = 125$, then $\sqrt[3]{125} = 5$. The cube-root symbol carries a small **3**: $\sqrt[3]{}$.

A **perfect cube** is a whole number used as a factor three times ($n \times n \times n = n^3$). Its **cube root** undoes that.



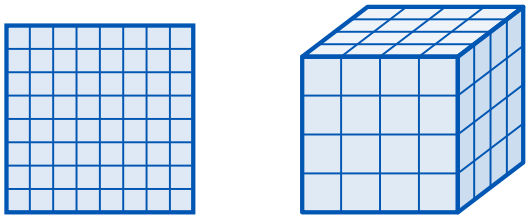
$2 \times 2 \times 2 = 8$ small cubes, so $2^3 = 8$ and $\sqrt[3]{8} = 2$. The edge is the cube root.

Perfect Cubes 1–10 — Reference Chart

These come up less often than squares, but knowing 1–10 makes cube roots instant.

n	n^3	n	n^3
1	1	6	216
2	8	7	343
3	27	8	512
4	64	9	729
5	125	10	1000

💡 **SQUARE ROOT vs CUBE ROOT:** √64 asks "what number times *itself* = 64?" → 8. ∛64 asks "what number used *three times* = 64?" → 4. Same number underneath, different answer — watch the little 3!



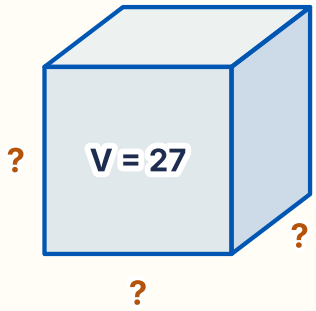
8 × 8 = 64
√64 = 8

4 × 4 × 4 = 64
∛64 = 4

64 tiles make an 8 × 8 square. 64 small cubes make a 4 × 4 × 4 cube.

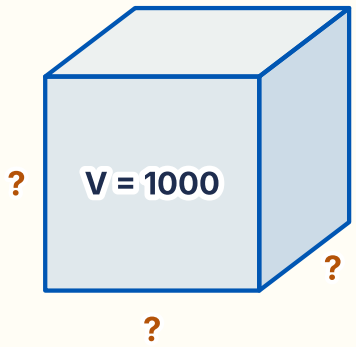
Quick Check

CALC **What is ∛27 ?**



? × ? × ? = 27. What edge length builds this cube?

MC **Which statement is TRUE?**



$\sqrt[3]{1000}$ is the edge of a cube with volume 1000: ? × ? × ? = 1000.

- ○ $\sqrt[3]{1000} = 100$
- ○ $\sqrt[3]{1000} = 10$
- ○ $\sqrt[3]{1000} = 3$
- ○ $\sqrt[3]{1000} = 500$

4 Estimating Square Roots

📌 **THE BIG IDEA:** When a number is NOT a perfect square, its square root is irrational — it never ends or repeats. We can't write it exactly, but we can *estimate* it. Two tools: a number line, and a step-by-step algorithm.

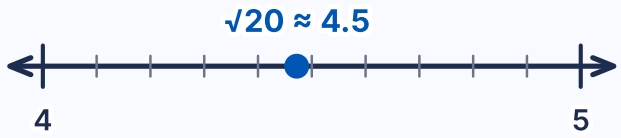
Method 1 — Place it on a Number Line

Great for "between which two whole numbers?" questions.

1. Find the two perfect squares your number lands **between**.
2. Take the √ of each — those are two back-to-back whole numbers. Your root lands between them.
3. Decide which one it's **closer** to: is your number nearer the lower or the upper perfect square?
4. Place it on the number line.

Example: place $\sqrt{20}$

16 < 20 < 25, so √20 is between 4 and 5. Since 20 is closer to 16, √20 sits a little below the middle — closer to 4.



16 < 20 < 25, so √20 is between 4 and 5. (Calculator: 4.47...)

Method 2 — The Estimating Algorithm (to the nearest tenth)

How far is your number into the gap?

Between the two perfect squares, find how far along your number sits. That fraction, added to the **smaller root**, is your estimate. Watch the colors: **gold = perfect squares**, **green = the roots**, **blue = your number**.

Step 1 — Trap your number between two perfect squares.

$$36 < 45 < 49$$

√36 = **6** · √49 = **7** → so √45 is between **6** and **7**. The smaller root is **6**.

Step 2 — Build the fraction: how far in is your number?

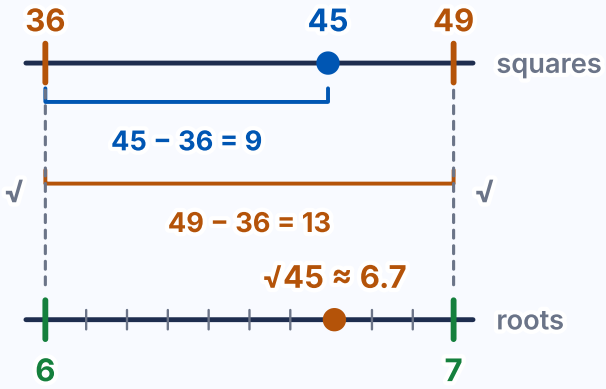
$$\frac{\text{your number} - \text{lower square}}{\text{upper square} - \text{lower square}} = \frac{45 - 36 = 9}{49 - 36 = 13} = 9 \div 13 \approx 0.7$$

The fraction is always *squares on top and bottom* — never the roots.

Step 3 — Add the fraction to the smaller root.

$$\sqrt{45} \approx 6 + 0.7 = 6.7$$

Calculator check: √45 = 6.708... ✓

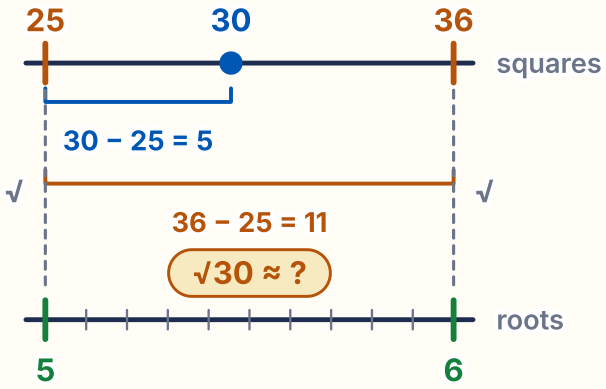


45 is **9 of the 13** steps from 36 to 49, so √45 is about 9 ÷ 13 ≈ 0.7 of the way from 6 to 7.

💡 **IN SHORT:** √n ≈ (smaller root) + (n − lower square) ÷ (upper square − lower square). This is an *estimate* — it can be off by about a tenth from a calculator, and that's fine.

⚠ **DON'T MIX THEM UP:** the number you **add** is a **root** (6). The numbers **inside the fraction** are the **perfect squares** (36 and 49). Roots to add, squares to subtract.

CALC **Estimate √30 to the nearest tenth using the algorithm.** (Hint: 25 < 30 < 36)



Use the two gaps to build your fraction, then add it to 5.

25 < 30 < 36 → roots 5 and 6. Fraction = (30 − 25) ÷ (36 − 25) = 5 ÷ 11 ≈ 0.45. So √30 ≈ 5 + 0.45 = 5.45 → **5.5**. (Calculator: 5.477…)

Negative Square Roots

📌 **JUST FLIP THE SIGN:** a negative in front puts your estimate on the other side of zero. Estimate the positive root first, then make it negative. Example: $-\sqrt{40}$ → $\sqrt{40} \approx 6.3$, so $-\sqrt{40} \approx -6.3$.



Same distance from 0, opposite sides: $\sqrt{40} \approx 6.3$ and $-\sqrt{40} \approx -6.3$.

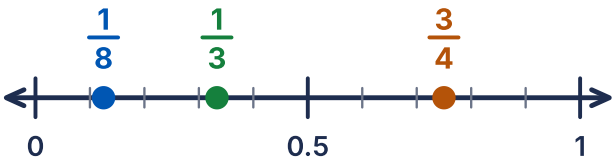
🧭 **WHICH METHOD?** Use the **number line** for "between which whole numbers." Use the **algorithm** when you need a decimal to the nearest tenth. Both start the same way: find the two perfect squares that surround your number.

5 Converting Fractions ↔ Decimals

Fraction → Decimal

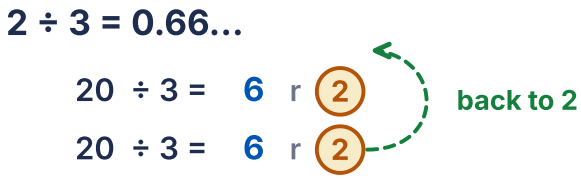
Divide the numerator by the denominator (top ÷ bottom).

Fraction	Division	Decimal
$\frac{3}{4}$	3 ÷ 4	_____
$\frac{1}{8}$	1 ÷ 8	_____
$\frac{1}{3}$	1 ÷ 3	_____



Each fraction is a point between 0 and 1. Dividing top ÷ bottom tells you its decimal name.

💡 **KEY INSIGHT:** Repeating decimals are _____ because they can be written as fractions. Non-repeating, non-terminating decimals (like π) are _____. Use a **bar** over the digits that repeat: $0.\overline{3}$ means 0.3333...



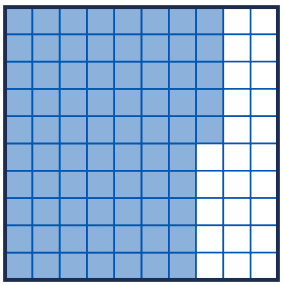
Same remainder again → the digits repeat

Why 2 ÷ 3 repeats: every step leaves remainder **2**, so the same digit (6) comes out forever.

Decimal → Fraction (terminating)

Read it, write it over its place value, then simplify — proof that every terminating decimal is **rational**.

Decimal	Read / write	Fraction (simplest form)
0.75	75 hundredths → $\frac{75}{100}$	= _____
0.6	6 tenths → $\frac{6}{10}$	= _____
0.125	125 thousandths → $\frac{125}{1000}$	= _____



0.75 = 75 of 100



0.6 = 6 of 10



0.125 = 125 thousandths

Read the place value: that is the fraction before you simplify.

Repeating Decimal → Fraction: The 9's Rule

📌 **THE 9's RULE:** The repeating digits go in the NUMERATOR. The DENOMINATOR is made of 9's — one 9 for each repeating digit.

Repeating digits	Denominator	Example	Simplified
1 digit	_____	$0.\overline{3} = \frac{3}{9}$	= _____
2 digits	_____	$0.\overline{27} = \frac{27}{99}$	= _____
3 digits	_____	$0.\overline{125} = \frac{125}{999}$	_____



Bracket one repeating block and count its digits. That count tells you how many 9s.

Worked Example — Convert $0.\overline{27}$ (0.272727…) to a fraction

Step 1 — Count the repeating digits: 2 digits repeat (the block "27").

Step 2 — Denominator = that many 9's: 2 digits → 99.

Step 3 — Numerator = the repeating block: $\frac{27}{99}$.

✓ **Step 4 — Simplify:** $\frac{27}{99} = \frac{3}{11}$.



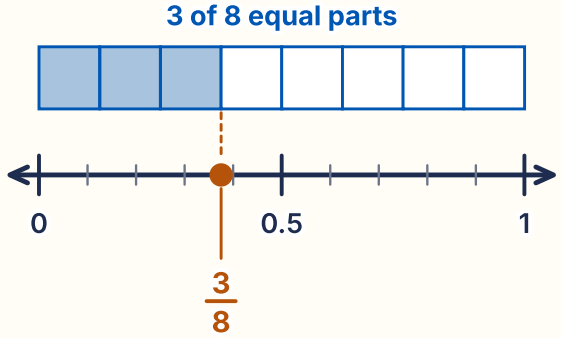
Check: 27 ÷ 99 gives 0.2727... back

Check the fraction: 27 ÷ 99 brings back the same remainder, 27, so the digits 2, 7 repeat.

⚠ **IMPORTANT:** Use the 9's rule when the repeat starts *right after* the decimal point, like 0.272727... Always simplify your fraction: $\frac{27}{99} = \frac{3}{11}$.

Quick Check

MC Which decimal equals the fraction $\frac{3}{8}$?



$\frac{3}{8}$ is 3 of 8 equal parts. Divide 3 ÷ 8 to name the point as a decimal.

- ○ 0.38
- ○ 0.375
- ○ 0.83
- ○ 0.625

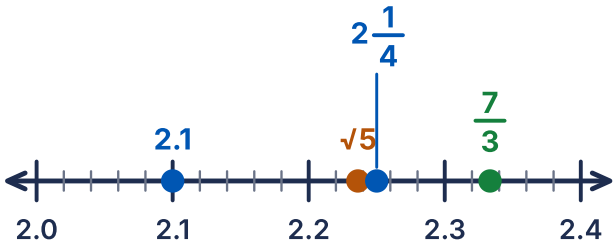
6 Ordering Numbers Least to Greatest

📌 **THE STRATEGY:** To compare a mix of fractions, decimals, mixed numbers, and roots — turn *everything* into a decimal first. Then ordering is easy.

Worked Example — Order $\sqrt{5}$, $2\frac{1}{4}$, $\frac{7}{3}$, 2.1

Number	Decimal	How?
$\sqrt{5}$	≈ _____	Estimating algorithm (between 4 and 9)
$2\frac{1}{4}$	= _____	2 + (1 ÷ 4) = 2.25
$\frac{7}{3}$	≈ _____	7 ÷ 3 = 2.333...
2.1	= _____	Already a decimal

Order from least to greatest: _____ < _____ < _____ < _____



Least → Greatest: **$2.1 < \sqrt{5} < 2\frac{1}{4} < \frac{7}{3}$**

💡 **MIXED NUMBERS:** turn them into a decimal — whole number + (numerator ÷ denominator). Example: $3\frac{2}{5}$ = 3 + (2 ÷ 5) = 3 + 0.4 = 3.4.

⚠ **WATCH OUT:**

- Give every number the same number of decimal places (3.5 → 3.500). Then 3.375 < 3.500 is easy to see.
- Estimate each root first (√5 ≈ 2.24).

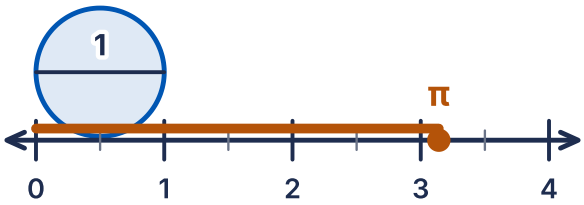
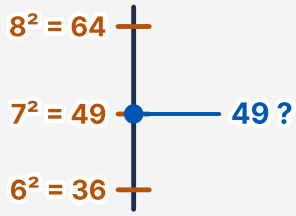
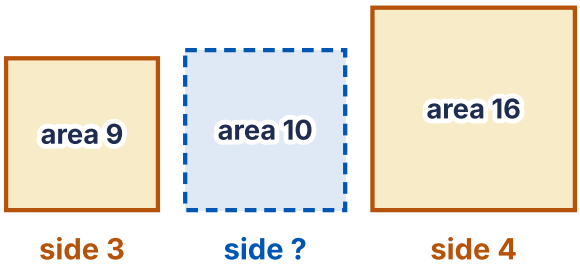
ERROR ANALYSIS To order 1.5, √2, $\frac{7}{4}$, √3, a student wrote $1.5 < \sqrt{2} < \sqrt{3} < \frac{7}{4}$. **Are they right?**

✏ PART 2 — Practice Problems

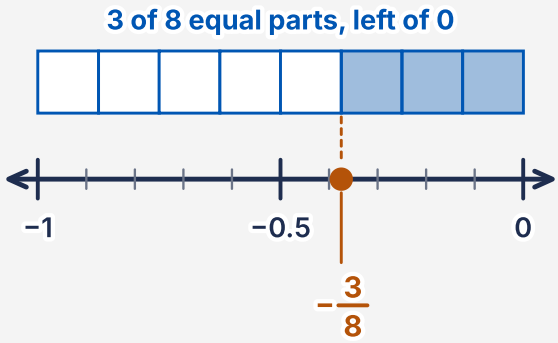
Show all work on your printed copy. Circle your final answer, then click to check on screen.

A Classify Each Number

Rational = can be written as a fraction (includes terminating & repeating decimals).
Irrational = non-repeating, non-terminating.

#	Classification
1. 0.333...  The 3 repeats forever.	_____
2. π  Roll the circle once: it goes π units π = 3.14159265358979...	_____
3. √49  Does 49 land on a perfect square?	_____
4. √10  10 is not a perfect square: its side is between 3 and 4.	_____

5. $-\frac{3}{8}$



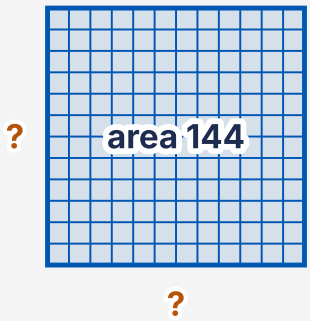
$-\frac{3}{8}$ is a fraction, 3 parts of 8 to the left of 0.

6. 0.1212...



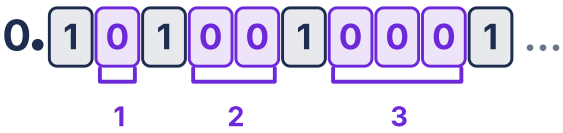
The block 12 repeats forever.

7. √144



144 tiles: count the side.

8. 0.101001...



The runs of 0s grow: 1, 2, 3, ... no block ever repeats.

9. $\frac{2}{7}$

2 ÷ 7 = 0.285714...

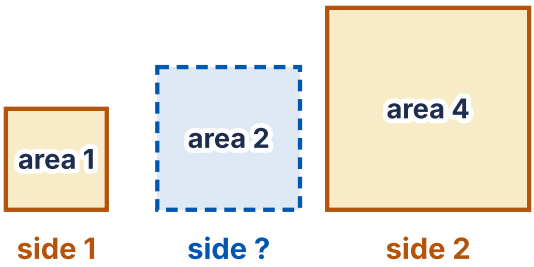
20 ÷ 7 = 2 r 6
60 ÷ 7 = 8 r 4
40 ÷ 7 = 5 r 5
50 ÷ 7 = 7 r 1
10 ÷ 7 = 1 r 3
30 ÷ 7 = 4 r 2

back to 2

Same remainder again → the digits repeat

_____

10. √2



2 is not a perfect square: its side is between 1 and 2.

_____

11. −√81



81 = 9 × 9. −√81 is the opposite of √81.

_____

12. 3.14

1s	10ths	100ths
3	1	4

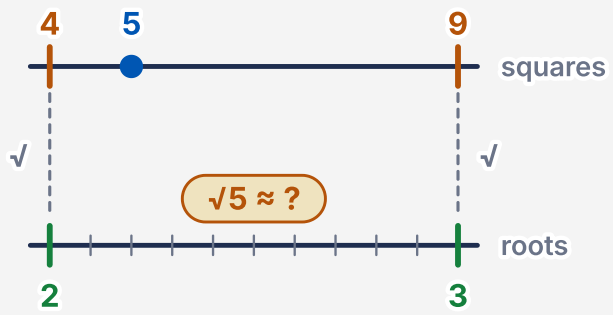
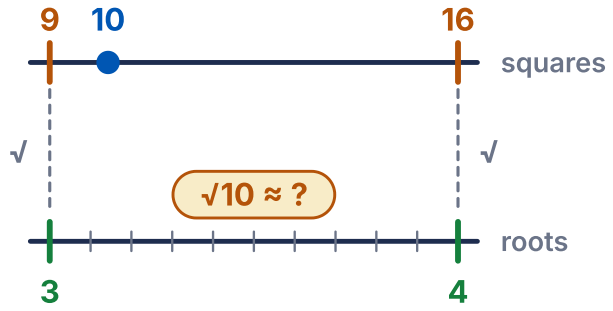
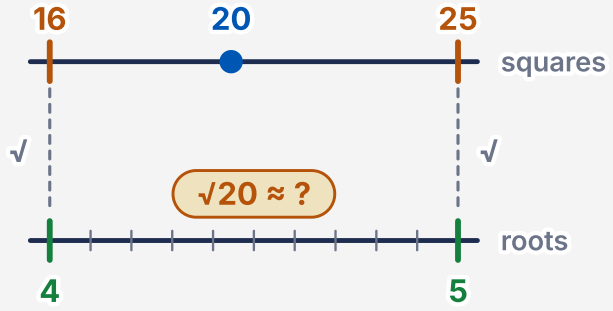
3 ones, 1 tenth, 4 hundredths

The decimal stops after the hundredths.

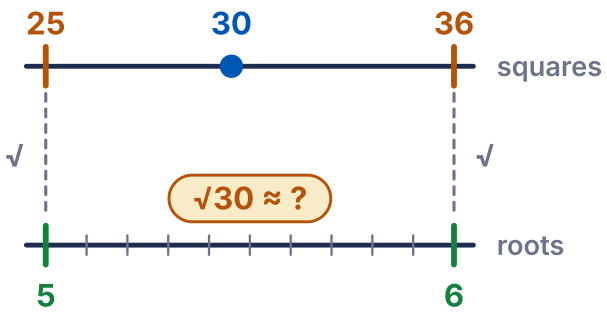
_____

B Estimate Each Square Root (nearest tenth)

Find the perfect squares just below and just above, then estimate. A leading – just flips the sign.

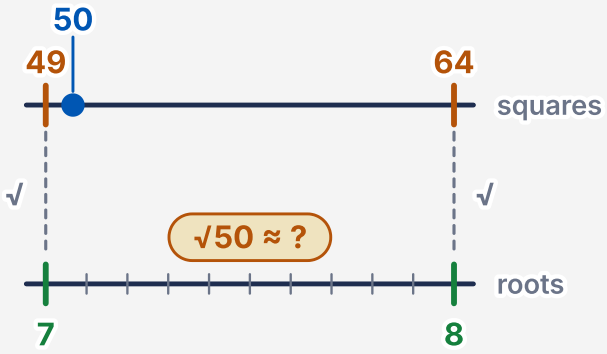
#	Estimate
1. √5 	≈ _____
2. √10 	≈ _____
3. √20 	≈ _____

4. √30



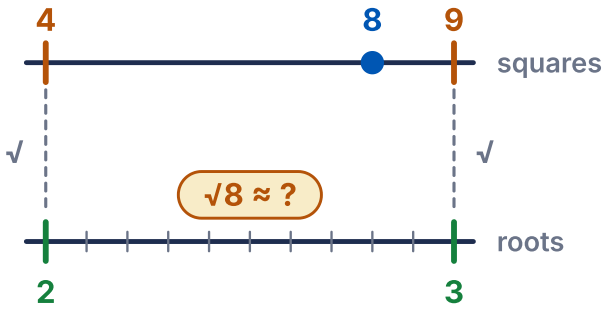
≈ _____

5. √50



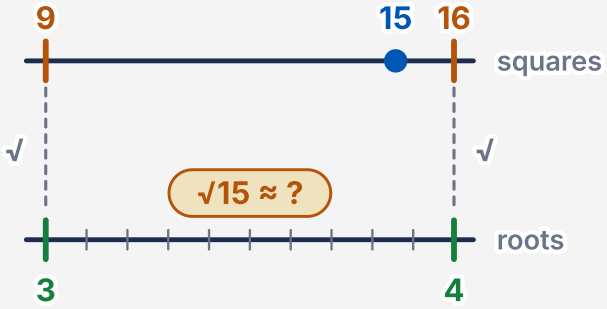
≈ _____

6. √8



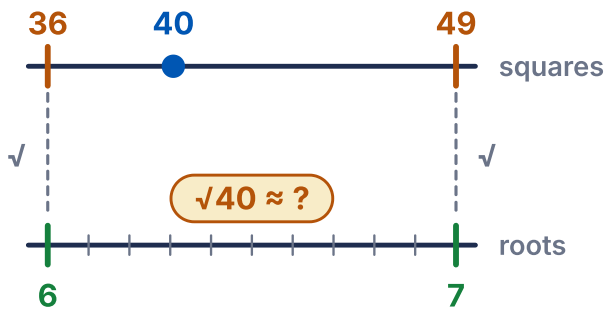
≈ _____

7. √15



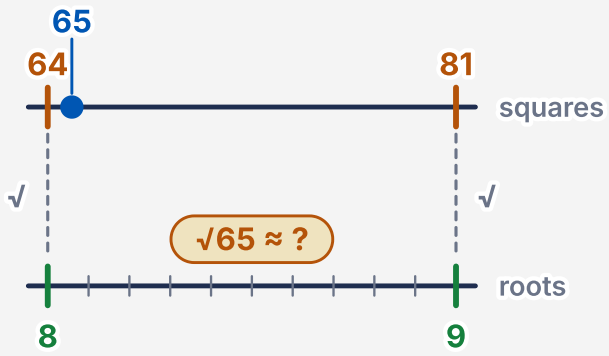
≈ _____

8. √40


36
40
49
squares
√
√
√40 ≈ ?
roots
6
7

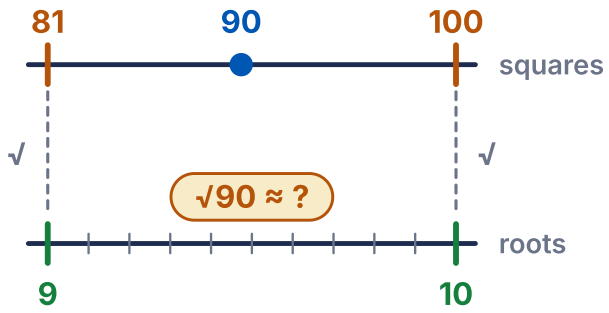
≈ _____

9. √65


65
64
81
squares
√
√
√65 ≈ ?
roots
8
9

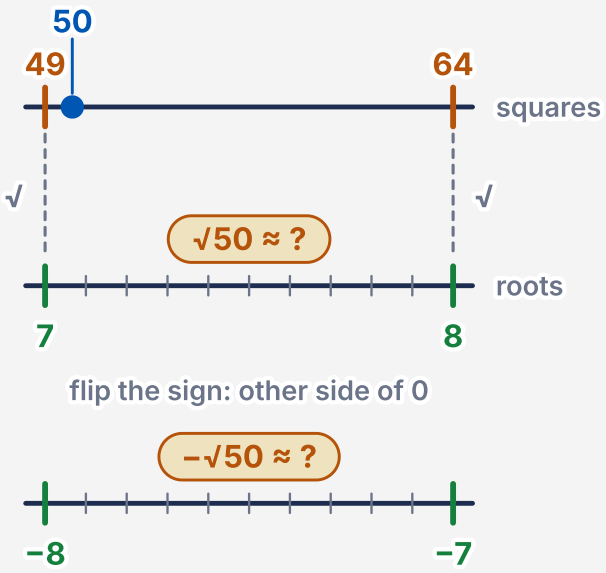
≈ _____

10. √90


81
90
100
squares
√
√
√90 ≈ ?
roots
9
10

≈ _____

11. $-\sqrt{50}$



flip the sign: other side of 0

≈ _____

C Order from Least to Greatest

Convert each number to a decimal, then arrange from least to greatest. Show your decimal conversions!

1. Order: π, √10, $\frac{10}{3}$, 3.5



Turn each number into a decimal, then place it on this line: π, √10, $\frac{10}{3}$, 3.5.

2. Order: 1.5, √2, $\frac{7}{4}$, √3



Turn each number into a decimal, then place it on this line: 1.5, √2, $\frac{7}{4}$, √3.

3. Order: √5, 2.5, $\frac{11}{5}$, √7



Turn each number into a decimal, then place it on this line: √5, 2.5, $\frac{11}{5}$, √7.

4. Order: 4.2, √18, $\frac{13}{3}$, √20



Turn each number into a decimal, then place it on this line: 4.2, √18, $\frac{13}{3}$, √20.

5. Order: -2.5, $-\sqrt{5}$, $-\frac{8}{3}$



Turn each number into a decimal, then place it on this line: -2.5, $-\sqrt{5}$, $-\frac{8}{3}$.

6. Order: $\sqrt{50}$, 7.1, $\frac{22}{3}$, $\sqrt{52}$



Turn each number into a decimal, then place it on this line: $\sqrt{50}$, 7.1, $\frac{22}{3}$, $\sqrt{52}$.

7. Order: $\frac{5}{6}$, $\sqrt{0.5}$, 0.9, $\frac{7}{8}$



Turn each number into a decimal, then place it on this line: $\frac{5}{6}$, $\sqrt{0.5}$, 0.9, $\frac{7}{8}$.

8. Order: $\sqrt{30}$, 5.5, $\frac{17}{3}$, $\sqrt{32}$



Turn each number into a decimal, then place it on this line: $\sqrt{30}$, 5.5, $\frac{17}{3}$, $\sqrt{32}$.